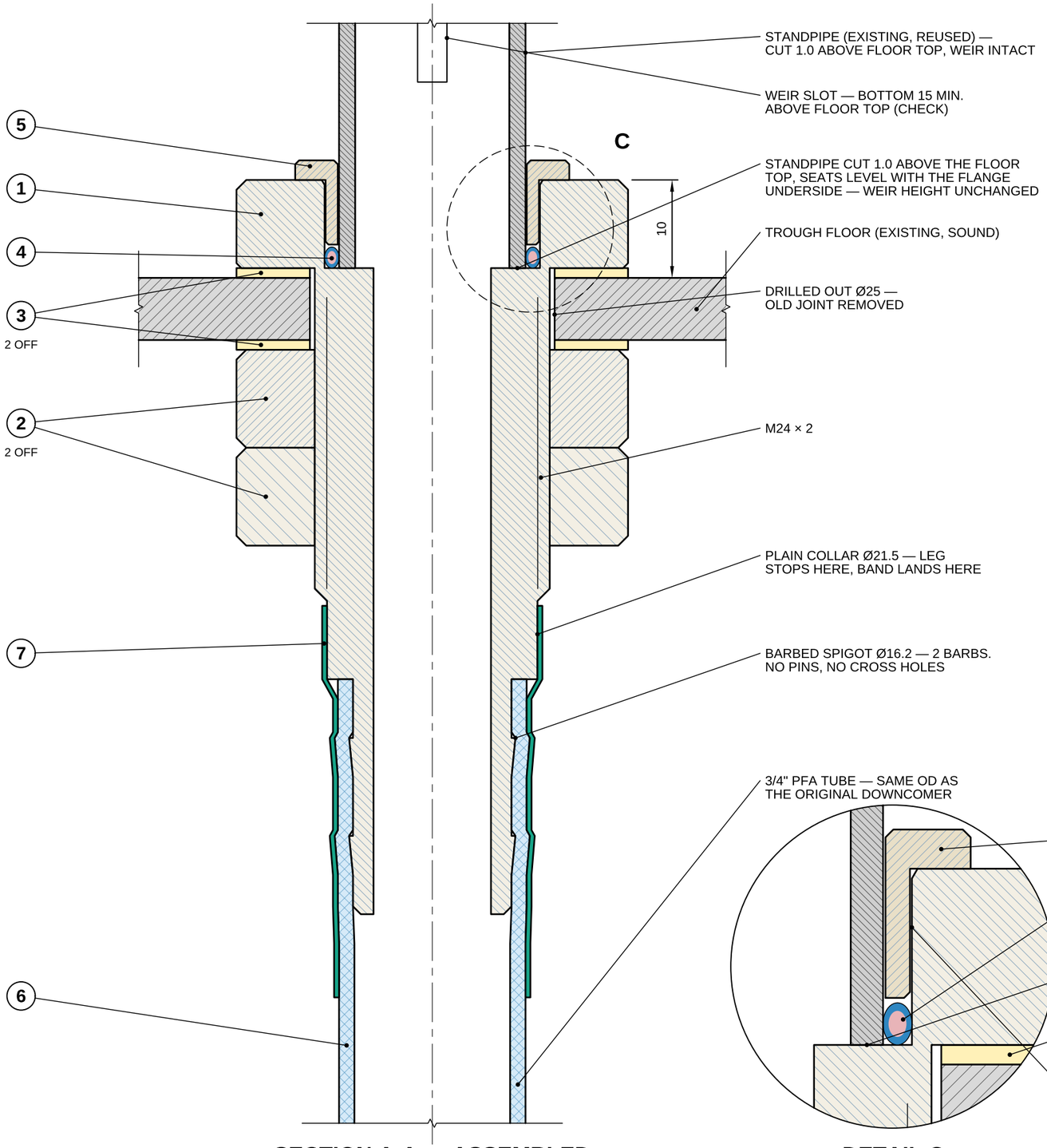
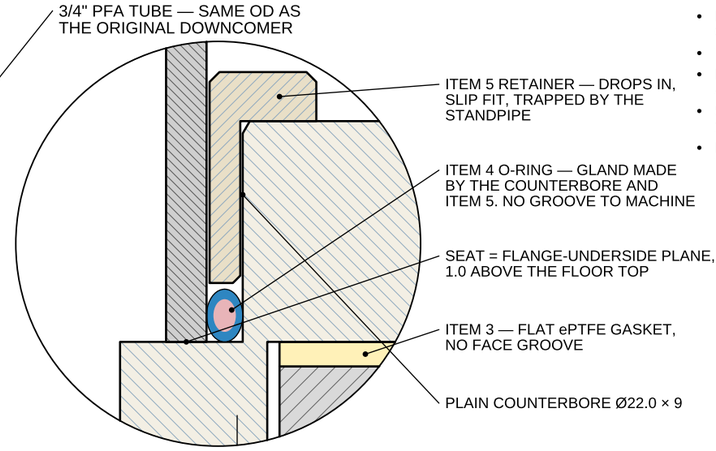


SECTION A-A — ASSEMBLED

2:1



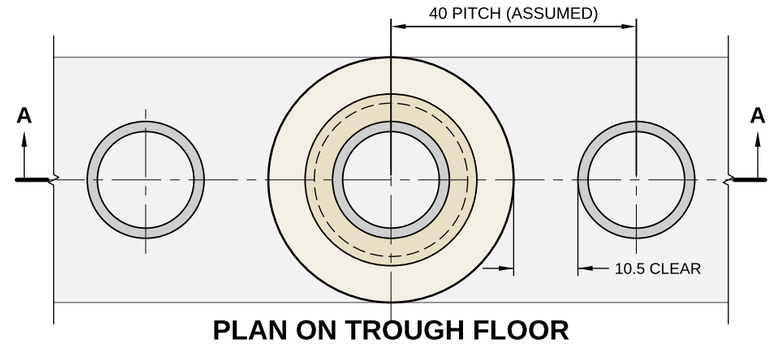
DETAIL C

4:1 (STANDPIPE SEAL)

PARTS LIST — PER DOWNCOMER

ITEM	QTY	DESCRIPTION	MATERIAL
1	1	INSERT — Ø40 FLANGE, Ø22 COUNTERBORE, M24 × 2 NECK, BARBED SPIGOT. DWG-DCINS-02.	PTFE, 25 % GLASS-FILLED
2	2	NUT M24 × 2, Ø40 × 10, PLAIN ROUND — CLAMP NUT AND JAM NUT, SAME PART. DWG-DCINS-02.	PTFE, 25 % GF
3	2	GASKET Ø40 × Ø25 × 1.5 — FLOOR TOP AND UNDERSIDE, SAME PART.	EXPANDED PTFE
4	1	O-RING 18.5 ID × 1.78 CS — STANDPIPE SEAL.	FEP-ENCAP., SILICONE CORE
5	1	SEAL RETAINER — Ø28 TOP HAT, DROP-IN. DWG-DCINS-02.	PTFE, 25 % GF
6	1	NEW LEG — TUBE 3/4" OD × 0.062" WALL (19.05 × 1.57), THERMOFORMED TO THE DOWNCOMER MARK, CUT TO THE DISCHARGE POINT.	PFA
7	1	JOINT BAND — HEAT-SHRINK 1" 1.6:1 (M23053/11-208 OR EQUAL), 40 LONG.	FEP
—	—	STANDPIPE — EXISTING, CUT 1.0 ABOVE THE FLOOR TOP AND REFITTED.	EXISTING

- GLASS-FILLED PTFE (1, 2)
- EXPANDED PTFE (3)
- PFA TUBE (6)
- EXISTING STANDPIPE
- GLASS-FILLED PTFE (5)
- O-RING, FEP JACKET (4)
- FEP BAND (7)



PLAN ON TROUGH FLOOR

1:1

DESIGN INTENT — REV B

- EASY TO MAKE — ITEM 1 IS TWO PLAIN LATHE OPERATIONS: NO INTERNAL GROOVE, NO FACE GROOVE, NO CROSS HOLES, NO MILLING. ITEMS 2 AND 5 ARE ONE OPERATION EACH.
- THE O-RING GLAND IS MADE BY A PLAIN COUNTERBORE AND A DROP-IN RETAINER (DETAIL C).
- FLAT ePTFE GASKETS BOTH SIDES OF THE FLOOR — THEY CONFORM TO A PITTED FLOOR. NUTS AND GASKETS ARE ONE PART EACH, USED TWICE.
- STANDPIPE CUT 1.0 ABOVE THE FLOOR TOP AND SEATED LEVEL WITH THE FLANGE UNDERSIDE, 1.0 UP — WEIR HEIGHT UNCHANGED.
- NEW LEG IS STOCK 3/4" PFA ON A BARBED SPIGOT UNDER AN FEP BAND. NOTHING METAL BELOW THE FLOOR.

GENERAL NOTES

1. DIMENSIONS IN mm.
2. EXISTING STANDPIPE 19.05 OD × 1.65 WT, FLOOR 6.35 AND 40 PITCH ARE ASSUMED — CONFIRM ON SITE BEFORE MACHINING.
3. FOR A DOWNCOMER WHOSE LEG IS GONE BUT WHOSE STANDPIPE, WEIR AND SURROUNDING FLOOR ARE SOUND.
4. ITEM 1, 2 AND 5 DETAILS ON DWG-DCINS-02. INSTALLATION ON DWG-DCINS-03.

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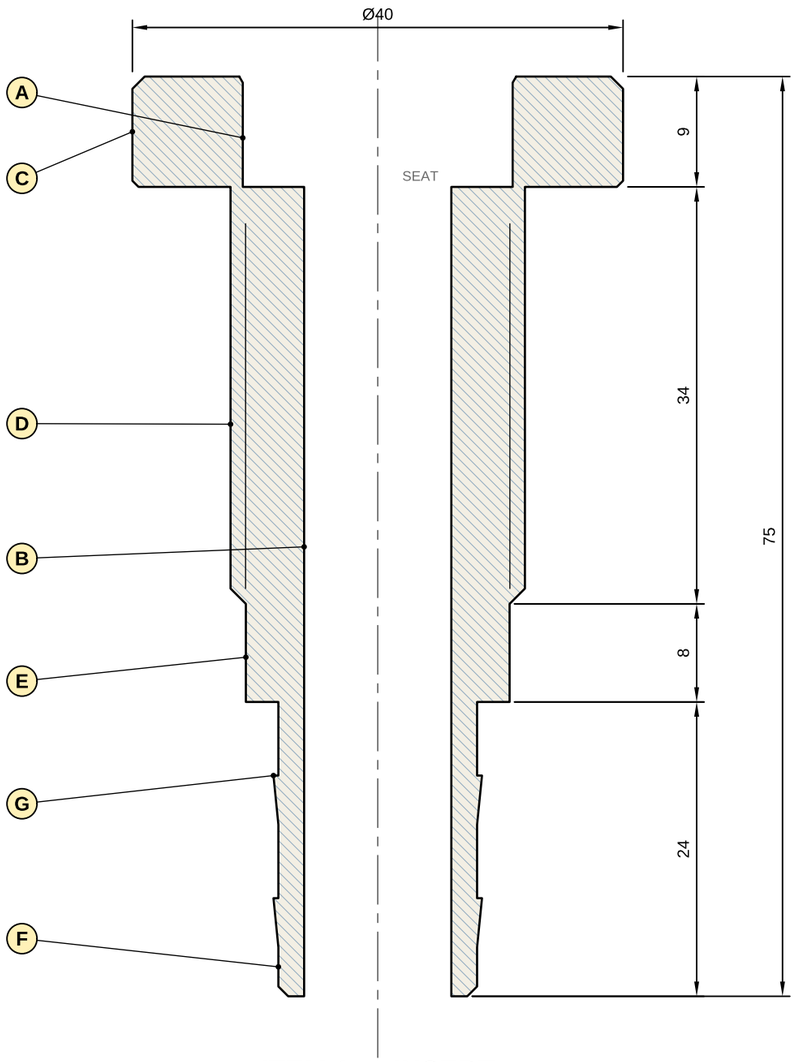
TITLE

DOWNCOMER RETROFIT INSERT

GENERAL ARRANGEMENT & PARTS LIST

CONCEPT — NOT FOR CONSTRUCTION

DESIGNED: J. YOUNG	DATE: 26/09/2026
DRAWING No.: DWG-DCINS-01	SCALE: AS SHOWN
SHEET: 1 OF 3	REV: B



ITEM 1 — INSERT

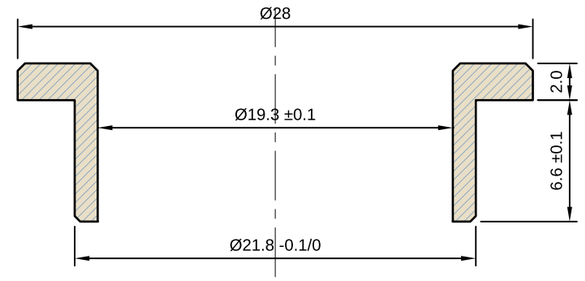
2:1 PTFE + 25 % GLASS, 1 OFF PER POSITION

ITEM 1 — TWO LATHE OPERATIONS, NOTHING ELSE

OP 10 Ø45 BAR, 90 OUT OF THE CHUCK. FACE. CENTRE-DRILL, DRILL Ø12 × 80 DEEP (B). TURN THE SPIGOT AND BOTH BARBS (F, G), THE COLLAR (E), THE THREAD OD, THE FLANGE UNDERSIDE AND THE FLANGE Ø40 (C). SINGLE-POINT THE M24 × 2 (D). CHAMFERS. PART OFF 1 LONG.

OP 20 REVERSE, HOLD ON THE Ø40 FLANGE IN SOFT JAWS. FACE TO 75.0 OVERALL. BORE THE Ø22.0 COUNTERBORE FLAT-BOTTOMED TO 9.0 DEEP (A). MOUTH CHAMFER 0.5 × 30°, TOP EDGE 1 × 45°.

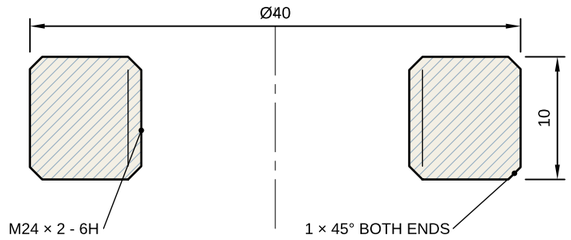
NO MILLING, NO CROSS HOLES, NO INTERNAL OR FACE GROOVES. ONE BORING BAR, ONE THREADING TOOL.



SPIGOT LENGTH SETS THE O-RING GLAND WIDTH (2.4).

ITEM 5 — SEAL RETAINER

3:1 PTFE + 25 % GLASS, 1 OFF. ONE OPERATION.



PLAIN ROUND — TIGHTEN WITH A STRAP WRENCH.

ITEM 2 — NUT

2:1 PTFE + 25 % GLASS, 2 OFF (CLAMP + JAM). ONE OPERATION.

STANDPIPE SEAL GLAND (ITEM 4)

O-RING	GLAND	SQUEEZE	FILL
18.5 × 1.78	Ø22.0 COUNTERBORE ON THE Ø19.05 STANDPIPE, 2.4 WIDE BETWEEN THE SEAT AND ITEM 5	17 % (11 % AT OD 18.85)	70 %

- FEP-ENCAPSULATED, SILICONE CORE. SIZE TO BE CONFIRMED WITH THE SUPPLIER.
- ENCAPSULATED RINGS BARELY STRETCH — WARM ITEM 4 IN HOT WATER BEFORE FITTING IT.
- STANDARD ITEM 1 SUITS A STANDPIPE OD OF 18.85–19.15 AT THE SEAL BAND. OUTSIDE THAT, BORE FEATURE A TO THE MEASURED OD + 2.95 — STILL OP 20 ONLY.

ITEM 1 — FEATURES

	FEATURE	SIZE / TOLERANCE	OP
A	COUNTERBORE	Ø22.0 +0.1/0 × 9.0 ±0.1 DEEP FROM THE TOP FACE. FLAT BOTTOM, LEVEL WITH THE FLANGE UNDERSIDE, IS THE STANDPIPE SEAT. MOUTH CHAMFER 0.5 × 30°.	20
B	BORE	Ø12.0 THROUGH.	10
C	FLANGE	Ø40 × 9. TOP EDGE 1 × 45°, UNDERSIDE EDGE 0.5 × 45°. UNDERSIDE FLAT, NO GROOVE.	10/20
D	THREAD	M24 × 2 - 6g × 34 FROM THE FLANGE UNDERSIDE. LAST 3 MAY BE INCOMPLETE (INSIDE THE FLOOR HOLE). 45° CHAMFER TO E.	10
E	COLLAR	Ø21.5 ±0.1 × 8 — PASSES THE NUT MINOR (Ø21.84). THE LEG STOPS ON ITS SHOULDER.	10
F	SPIGOT	Ø16.2 ±0.1 × 24. END CHAMFER 0.8 × 45°.	10
G	BARBS	2 OFF Ø17.0 ±0.1, SHARP FACE TOWARD E, AT 6 AND 16 FROM E. 4 LONG RAMP.	10

ITEMS 2 AND 5 — ONE OPERATION EACH

ITEM 2 Ø45 BAR. FACE. DRILL Ø22.0, THREAD M24 × 2 (TAP OR SINGLE-POINT), CHAMFER THE THREAD BOTH ENDS. TURN Ø40, CHAMFER. PART OFF AT 10, DEBURR. SAME PART FOR THE CLAMP AND THE JAM NUT.

ITEM 5 Ø30 BAR. FACE. DRILL AND BORE Ø19.3. TURN THE Ø21.8 SPIGOT 6.6 LONG AND THE Ø28 FLANGE. CHAMFERS. PART OFF AT 8.6.

MANUFACTURING NOTES

1. MATERIAL: PTFE + 25 % GLASS FIBRE, MOULDED ROD FROM VIRGIN RESIN — Ø45 FOR ITEMS 1 AND 2, Ø30 FOR ITEM 5. CERTIFICATE OF CONFORMANCE PER BATCH.
2. STRESS-RELIEVE (ANNEAL) BLANKS BEFORE FINISH CUTS. CARBIDE TOOLING (THE GLASS IS ABRASIVE), SHARP, POSITIVE RAKE, LIGHT FINISHING CUTS, NO COOLANT OR WATER-FREE COOLANT.
3. MEASURE AT 20–23 °C — PTFE CHANGES SIZE THROUGH 19 °C.
4. ONLY A, E, F, G AND THE ITEM 5 SPIGOT AND BORE CARRY TIGHT TOLERANCES. EVERYTHING ELSE ±0.3.
5. SEAL SURFACES (A, SEAT, FLANGE UNDERSIDE, ITEM 5 SPIGOT): Ra 1.6, NO BURRS.
6. RUN A NUT ONTO EVERY ITEM 1 BY HAND BEFORE DISPATCH.
7. MARK PART No., BATCH AND YEAR ON THE FLANGE TOP OUTSIDE Ø28 — ENGRAVE 0.2 DEEP.
8. ITEM 3: PUNCH FROM 1.5 EXPANDED PTFE SHEET, Ø40 × Ø25.
9. KIT PER POSITION: 1, 2 × 2, 3 × 2, 4, 5 — ITEMS 4 AND 5 CAN BE FITTED INTO ITEM 1 AT THE BENCH. ITEMS 6 AND 7 FROM STOCK ON SITE. CLEAN, DRY, OIL-FREE.

GENERAL NOTES

1. DIMENSIONS IN mm. GENERAL TOLERANCE ±0.3 UNLESS SHOWN.
2. SIZES SUIT A 19.05 OD STANDPIPE (ASSUMED). IF THE MEASURED OD DIFFERS, ONLY FEATURE A CHANGES.
3. READ WITH DWG-DCINS-01 (ASSEMBLY) AND DWG-DCINS-03 (INSTALLATION).

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TITLE

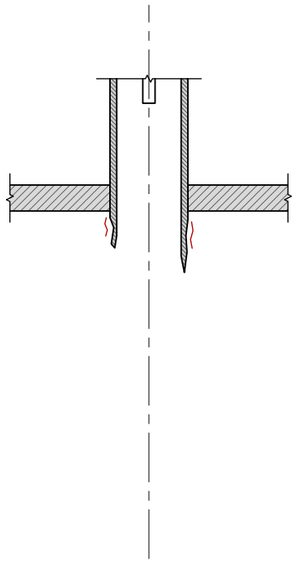
DOWNCOMER RETROFIT INSERT

ITEM 1, 2 & 5 — MACHINING DETAILS

CONCEPT — NOT FOR CONSTRUCTION

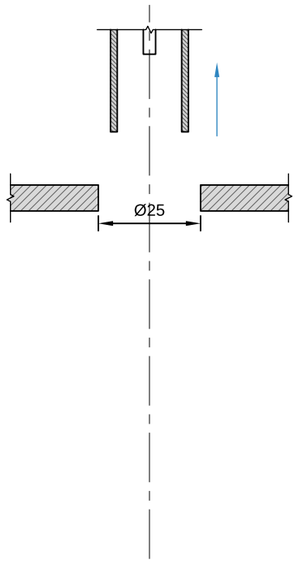
DESIGNED J. YOUNG	DATE 26/09/2026
DRAWING No. DWG-DCINS-02	SCALE AS SHOWN
SHEET 2 OF 3	REV B

INSTALLATION SEQUENCE — ONE POSITION (1:1.5)



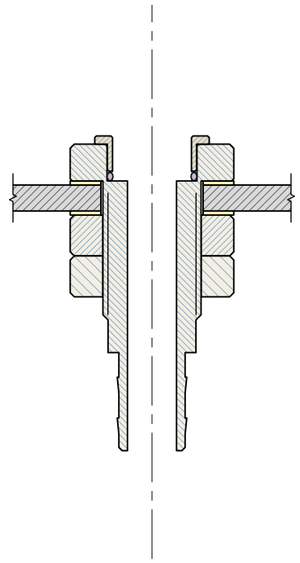
a AS FOUND

LEG EATEN THROUGH AT OR JUST BELOW THE FLOOR. STANDPIPE, WEIR AND FLOOR STILL SOUND.



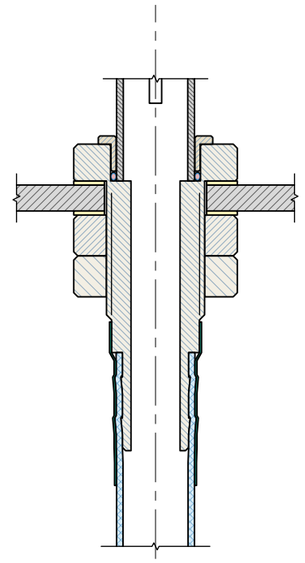
b CUT + DRILL

STANDPIPE CUT 1.0 ABOVE THE FLOOR TOP AND LIFTED OUT. Ø25 DRILLED ON THE STUB CENTRE — OLD JOINT GONE.



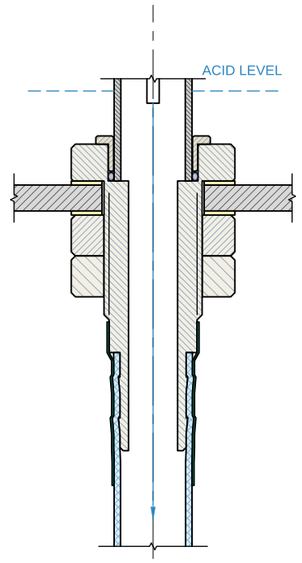
c INSERT + CLAMP

INSERT, WITH O-RING AND RETAINER FITTED AT THE BENCH, IN FROM ABOVE ON A GASKET. FROM BELOW: GASKET, NUT, JAM NUT.



d STANDPIPE + LEG

STANDPIPE PUSHED DOWN THROUGH THE RETAINER ONTO ITS SEAT. PFA LEG PUSHED OVER THE BARBS, BAND SHRUNK OVER.



e TEST

PACKING BACK BY HAND. CIRCULATE — LEVEL IN THE WEIR MATCHES ITS NEIGHBOURS.

1. WHICH REPAIR?

CONDITION FOUND	REPAIR	DRAWING
LEG DAMAGED BELOW THE FLOOR, MOSTLY FROM OUTSIDE. JOINT TO THE FLOOR STILL SOUND.	ENCAPSULATION SLEEVE	DWG-DCSLV-01, 02
LEG GONE AT OR THROUGH THE JOINT, OR THE JOINT LEAKS. STANDPIPE, WEIR AND FLOOR AROUND IT STILL SOUND.	**RETROFIT INSERT (THIS SET)**	**DWG-DCINS-01–03**
FLOOR THINNED OR HOLED AROUND THE POSITION — WILL NOT TAKE A CLAMP.	BULKHEAD INSERT	DWG-DCBHD-01, 02

2. DESIGN BASIS

- WEIR UNCHANGED — THE STANDPIPE IS CUT 1.0 ABOVE THE FLOOR TOP AND SEATS LEVEL WITH THE FLANGE UNDERSIDE, WHICH SITS 1.0 UP ON ITS GASKET. SLOT HEIGHTS AND FLOW STAY MATCHED TO THE NEIGHBOURS.
- LOAD PATH — FLANGE ON THE FLOOR TOP AND NUT ON THE UNDERSIDE, EACH THROUGH A FLAT ePTFE GASKET. THE FLOOR CARRIES THE CLAMP; THE INSERT CARRIES ONLY THE STANDPIPE AND THE LEG.
- SEALING — THE TOP GASKET STOPS ACID BYPASSING THE WEIR; O-RING 4 SEALS THE STANDPIPE IN A GLAND MADE BY THE COUNTERBORE AND ITEM 5. ONLY THE TROUGH LEVEL (TENS OF mm) ACTS ON EITHER SEAL.
- MADE ON A LATHE — ITEM 1 IN TWO PLAIN OPERATIONS, ITEMS 2 AND 5 IN ONE EACH. NO GROOVES, CROSS HOLES OR MILLING.
- NO METAL ADDED, NO WELDING, NO EXPANDING. EVERYTHING BELOW THE FLOOR IS FLUOROPOLYMER.
- SERVICE — H2SO4 98–99.5 %, UPPER-STAGE ACID 70–100 °C TYPICAL. PFA AND FEP RATED WELL ABOVE THIS; CREEP OF GLASS-FILLED PTFE UNDER CLAMP LOAD IS THE LIMIT TO PROVE.

3. INSTALLATION

1. SURVEY: MARK AND PHOTOS. STANDPIPE SOUND, WEIR INTACT, OD AT THE SEAL BAND (1–12 ABOVE THE FLOOR) 18.85–19.15. UT THE FLOOR OUT TO Ø44. RECORD THE DISCHARGE POINT AND SLOT ORIENTATION.
2. CONFIRM THE WEIR SLOT BOTTOM IS 15 OR MORE ABOVE THE FLOOR TOP — ITEM 5 STANDS 12 PROUD.
3. BENCH: FIT ITEM 4 (WARMED) AND ITEM 5 INTO ITEM 1. THERMOFORM ITEM 6 TO THE MARK, 40 STRAIGHT AT THE TOP.
4. CUT THE STANDPIPE 1.0 ABOVE THE FLOOR TOP (INTERNAL TUBE CUTTER, KERF INTO THE STUB). LIFT OUT, TAG, CHAMFER THE CUT END 0.5 × 30°, DEBURR.
5. DRILL Ø25 ON THE STUB CENTRE FROM INSIDE THE TROUGH (PILOTED ANNULAR CUTTER) — STUB AND OLD JOINT COME OUT WITH THE CORE. CLEAR THE OLD LEG FROM BELOW. DEBURR BOTH SIDES.
6. DRY-CLEAN BOTH FACES OF THE FLOOR OUT TO Ø44. NO WATER.
7. ITEM 3 ON THE FLOOR TOP, ITEM 1 IN FROM ABOVE. FROM BELOW: ITEM 3, THEN ITEM 2 — HAND TIGHT UNTIL THE FLANGE SEATS, THEN 1/4 TURN WITH A STRAP WRENCH. SECOND ITEM 2 JAMMED AGAINST THE FIRST.
8. PUSH THE STANDPIPE DOWN THROUGH ITEMS 5 AND 4, TWISTING, UNTIL IT SEATS. SLOTS FACING AS RECORDED.
9. FROM BELOW: WARM THE END OF ITEM 6 (APPROX. 100 °C) AND PUSH IT OVER THE BARBS TO THE COLLAR, AIMED AT THE DISCHARGE POINT. ITEM 7 OVER THE JOINT, RECOVERED TOP-DOWN (AS DWG-DCSLV-02).
10. REPLACE FINAL PACKING BY HAND. CIRCULATE ACID AND CHECK AS SECTION 5.

4. FIT CHECK (ASSUMED DIMENSIONS)

- M24 NECK IN Ø25 HOLE — 0.5 RADIAL CLEARANCE.
- BELOW THE FLOOR: 1 + 10 + 10 = 21 ON 25.4 OF FULL THREAD — 4 SPARE.
- NUT MINOR Ø21.84 PASSES THE Ø21.5 COLLAR AND THE Ø17 BARBS.
- FLANGE TO NEIGHBOUR TUBE AT 40 PITCH: 10.5 CLEAR.
- ITEM 5 TOP 12 ABOVE THE FLOOR TOP; SLOT BOTTOM 15 MIN. — 3 CLEAR.
- HEAT: THE GRIP SLACKENS APPROX. 0.05 AT 100 °C — PROVE IN THE HOT SOAK.

5. INSPECTION & ACCEPTANCE

- FLANGE SEATED, TOP GASKET EVEN ALL ROUND. JAM NUT TIGHT.
- STANDPIPE HOME; TOP AND SLOTS LEVEL WITH NEIGHBOURS ±1. DISCHARGE POINT ±10.
- LEG HARD AGAINST THE COLLAR; BAND FULLY RECOVERED, NO LIFTED EDGE.
- CIRCULATION TEST: WEIR LEVEL MATCHES THE NEIGHBOURS. NO DRIP AT THE FLANGE, NUT OR LEG JOINT.

6. LIMITS OF USE (PROVISIONAL)

- FLOOR SOUND FROM Ø25 OUT TO Ø44 ON BOTH FACES. THINNED OR PITTED — USE DWG-DCBHD.
- PITCH 32 OR MORE; THE Ø40 NUTS CLEAR OF THE NEIGHBOURS' LEGS BELOW THE FLOOR.
- STANDPIPE OD AT THE SEAL BAND 18.85–19.15, OR BORE FEATURE A TO SUIT.
- ACCESS BELOW THE FLOOR FOR THE NUTS AND LEG (FINAL PACKING OUT LOCALLY).

7. TO BE CONFIRMED

- STANDPIPE OD AND WALL, FLOOR THICKNESS, PITCH, SLOT HEIGHTS — SITE OR OEM DRAWING.
- DESIGN FLOW PER DOWNCOMER THROUGH THE Ø12 BORE (58 % OF THE ORIGINAL AREA; THE LEG IS FULL BORE).
- GLASS-FILLED PTFE GRADE, GASKET COMPRESSION AND CLAMP RETENTION HOT — ACID SOAK ON A SCRAP OFFCUT.
- BARB PULL-OFF WITH WARMED 3/4" PFA — BENCH TEST.
- CUTTER FOR THE TROUGH ALLOY; INTERNAL TUBE CUTTER FOR A 15.75 BORE; O-RING SUPPLY.

GENERAL NOTES

1. READ WITH DWG-DCINS-01 AND -02. ITEM NUMBERS AS THE SHEET 1 PARTS LIST.
2. SEQUENCE SKETCHES 1:1.5, SECTIONED ON THE DOWNCOMER CENTRE.
3. LIMITS OF USE ARE PROVISIONAL UNTIL PROVEN BY TRIAL.

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TITLE

DOWNCOMER RETROFIT INSERT

SELECTION, LIMITS & INSTALLATION

CONCEPT — NOT FOR CONSTRUCTION

DESIGNED **J. YOUNG**	DATE **26/09/2026**
DRAWING No. **DWG-DCINS-03**	SCALE **1:1.5 / NTS**
SHEET **3 OF 3**	REV **B**